TL-04

POWER & BATTERY CALCULATOR

Plan essential electricity needs, estimate backup runtime, and document safe charging options before an interruption.

PURPOSE

This worksheet helps you identify which devices matter most, estimate their energy use, compare that demand with available backup power, and plan safe recharging. Equipment ratings vary; use the labels and manuals for your actual devices.

HOUSEHOLD / SITE PROFILE

Household / Site Name: _____

Planning Start Date: _____ | People Supported: _____

CHOOSE A PLANNING PERIOD

■ 24 Hours

■ 72 Hours

■ 7 Days

■ 14 Days

■ Other: _____

■ Review Date: _____

POWER PRIORITIES

■ Medical equipment / medically necessary devices

■ Emergency communications (Radios / NOAA)

■ Lighting (Flashlights / Lanterns)

■ Phone charging

■ Refrigeration for essential items

■ Temperature safety / fans

■ Water pump or essential controls

■ Work, accessibility, or care needs

IMPORTANT SAFETY NOTE

This is a planning aid, not an electrical design. Never modify wiring or connect a generator to household wiring except through equipment installed by a qualified professional. Keep generators outdoors and far from doors, windows, and vents as directed by the manufacturer and current safety guidance. Never operate fuel-burning equipment indoors or in enclosed spaces.

1. DEVICE LOAD INVENTORY

List the devices you intend to power. Use the device label or manual for watts (W), and record how many hours per day it will run. For each device: daily watt-hours (Wh) = watts x hours per day x quantity.

Device / Purpose	Watts (W)	Hours/Day	Qty	Daily Wh
_____	_____	_____	____	_____
_____	_____	_____	____	_____
_____	_____	_____	____	_____
_____	_____	_____	____	_____
_____	_____	_____	____	_____
_____	_____	_____	____	_____
_____	_____	_____	____	_____
_____	_____	_____	____	_____

ESTIMATE DAILY ENERGY TOTALS

Daily Energy Total (Sum of all Daily Wh above):	_____ Wh
Essential-Only Total (Critical loads only):	_____ Wh
Planning Days (Number of days from page 1):	_____ Days
Estimated Total Energy Matrix (Daily Total x Days):	_____ Wh

ESSENTIAL / OPTIONAL CLASSIFICATION SYSTEM

■ **Critical:** Cannot safely go without under any condition.

■ **Important:** Reduce use or cycle tool if necessary.

■ **Optional:** Suspend entirely during power shortages.

■ **Medical device requirements verified**

■ **Device startup / surge requirements checked**

■ **Manufacturer instructions available**

2. BATTERY & BACKUP ESTIMATE

Battery labels show watt-hours (Wh) or amp-hours (Ah) and voltage (V). Compare capacities in watt-hours. If only Ah and V are known, approximate: Wh = Ah x V. Check manufacturer guides for true discharge limits.

Unit / Battery Model	Capacity (Wh/Ah)	Voltage (V)	Usable Wh	Condition / Date
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

RUNTIME WORKSHEET

Estimated Runtime = Usable Battery Wh ÷ Device Watts. Real results are lower due to conversion losses.

Device Load	Watts	Usable Battery Wh	Estimated Hours	Operational Notes
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

BACKUP SOURCES INVENTORY AVAILABLE

- ■ USB portable power banks
- ■ Rechargeable power tool battery packs
- ■ Dedicated portable power stations
- ■ Portable solar panels / controller
- ■ Vehicle charging loop (ventilated use only)
- ■ Professionally installed backup system

3. RECHARGING & SAFE OPERATION

Backup power resources stretch furthest when loads are strictly minimized and recharge windows are pre-planned.

Source / Charger	Compatible Unit	Expected Charge Time	Safe Location / Notes
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

LOAD-MANAGEMENT RULES

■ Charge all phones and packs before expected outages.

■ Enable ultra low-power or battery-saver mode.

■ Turn off all nonessential lights and electronics.

■ Unplug phantom or standby loads immediately.

■ Keep required medical equipment strictly prioritized.

■ Keep essential charging cables and adapters together.

■ Monitor battery state and remaining runtimes hourly.

■ Know where local utility updates are broadcast.

FUEL-POWERED EQUIPMENT REGULATIONS

■ **Operate strictly outdoors**, far from windows and vents.

■ Follow manufacturer alignment and grounding specs.

■ Ensure carbon-monoxide alarms are active indoors.

■ Store all fuel safely in approved containers legally.

■ **Allow equipment to cool completely** before refueling.

■ Keep electronic connections completely dry and protected.

■ Never backfeed household outlets or improvise wiring.

■ Follow local fire and electrical emergency instructions.

SITE OPERATIONS LOCATION METRICS

Outdoor Operating Area / Clearance Plan / Assigned Operator Line:

_____

_____

4. ACTION PLAN & VERIFICATION

Turn estimates into direct, manageable actions. Recheck this log whenever hardware or household needs alter.

TOP FIVE IMMEDIATE ACTIONS REQUIRED

Priority	Action / Hardware Item	Target Capacity	Assigned User	Due Date
1	_____	_____	_____	_____
2	_____	_____	_____	_____
3	_____	_____	_____	_____
4	_____	_____	_____	_____
5	_____	_____	_____	_____

POWER READINESS SNAPSHOT LOG

Sizing Sector	Have Stored	Deficit Gap	Next Procurement Target
Critical Illumination	_____ Wh	_____ Wh	_____
Device Communications	_____ Wh	_____ Wh	_____
Essential Medical Backup	_____ Wh	_____ Wh	_____
Auxiliary Engine Fuel	_____ Gal	_____ Gal	_____

HOUSEHOLD SPECIFIC MEDICAL OR ACCESSIBILITY OVERRIDES

_____
_____
_____
_____

CRITICAL LIFE SAFETY REGULATION AND AUTHORITATIVE REFERENCE REGISTER

LEGAL AND COMPLIANCE DISCLAIMER NOTICE: THIS WORKSHEET IS AN INDEPENDENT PLANNING AID AND RESILIENCE ENGINE METRIC MATRIX BLUEPRINT, NOT A SUBSTITUTE FOR PROFESSIONAL ELECTRICAL SERVICES, LICENSED ENGINEERING DESIGN, ACCREDITED COMPLIANCE CODES, OR DIRECT CLINICAL SAFETY SCHEMES. HOUSEHOLD ELECTRICAL DEMANDS, BACKUP runtime PARAMETERS, AND RECHARGE HOUSING ENVIRONMENTS ROUTINELY ALTER BY HARDWARE AGE AND WEATHER SEVERITY. IN AN ACTIVE INFRASTRUCTURE FAILURE OR IMMINENT DEFICIT THREAT, THE USER IS ORDERED TO Refer DIRECTLY TO MANDATED SAFETY COMMAND PROCEDURES DICTATED BY RESPONSIBLE PUBLIC PROTECTION AUTHORITIES, INCLUDING THE U.S. CONSUMER PRODUCT SAFETY COMMISSION (CPSC), THE FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA), THE CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC), MUNICIPAL CODE ENFORCERS, OR YOUR REGIONAL POWER INFRASTRUCTURE CONCERN LEADERS.

FIELD TECHNICAL SOURCES AND STRATEGIC DATA REFERENCES:

- U.S. Consumer Product Safety Commission (CPSC) - Portable Generator Carbon Monoxide Hazard Codes
- Federal Emergency Management Agency (FEMA) / Ready.gov - Power Outage Mitigation Infrastructure Standards
- Centers for Disease Control and Prevention (CDC) - Residential Carbon Monoxide Poisoning Prevention Matrix

Prepared for Household	Last System Review	Next Scheduled Review
_____	_____	_____